

MEFISTO is a novel computational framework that extends traditional factor analysis to address the challenges posed by multimodal datasets with inherent temporal or spatial structure. Below is a detailed summary of the paper, including the core concept, methodology, experiments, and how the method works.

1. Concept and Motivation

- **Background:**
Factor analysis is widely used in genome biology for dimensionality reduction and uncovering the major axes of variability in high-throughput data. However, standard factor analysis methods assume that samples are independent, an assumption that does not hold in experiments where data are collected over time or across spatial locations.
 - **Motivation for MEFISTO:**
Many modern biological studies—such as developmental atlases, longitudinal microbiome profiling, and spatial transcriptomics—generate datasets where samples have known relationships given by continuous covariates (e.g., developmental time, spatial coordinates). MEFISTO was developed to explicitly model these continuous dependencies and thereby improve the interpretability and accuracy of the latent factors extracted from such data.
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2. Methodology

Data Input and Structure

- **Multimodal Data:**
MEFISTO accepts high-dimensional data matrices collected from one or multiple “views” (different omics or measurement types) and organized by “groups” (e.g., individuals, species, or conditions).
- **Continuous Covariates:**
Each sample is annotated with one or more continuous covariates (such as time points or spatial coordinates) that capture the inherent structure in the data.

Model Framework

- **Factorization Approach:**
The model decomposes the input data tensor (with dimensions corresponding to samples, features, views, and groups) into two main components:
 - A latent factor matrix $\mathbf{Z}^g$ for each group g (with K factors), and
 - A set of weight matrices $\mathbf{W}^m$ for each view m .These are combined as:

$$\mathbf{Y}_{m,g} = \mathbf{Z}^g \mathbf{W}^m \mathbf{T} + \epsilon_{m,g} \quad \mathbf{Y}^{m,g} = \mathbf{Z}^g \mathbf{W}^m \mathbf{T} + \epsilon^{m,g}$$

- **Gaussian Process Prior:**
Each latent factor is modeled as a function of the continuous covariate using a

Gaussian process (GP) prior. In other words, each factor value z_{nk}^g is generated as:

$$z_{nk}^g = f_k(c_{ng}) + \eta_{nk}^g, f_k \sim \text{GP}(0, \kappa_k) \quad z_{nk}^g = f_k(\mathbf{c}_{n^g}) + \eta_{nk}^g, \quad f_k \sim \text{GP}(0, \kappa_k)$$

- The covariance function κ_k is designed to capture both the similarity between groups (through a low-rank covariance function κ_k^G) and the smooth variation along the covariate (via a squared exponential kernel κ_k^C): $\kappa_k(c_{ng}, c_{lh}) = \kappa_k^G(g, h) \kappa_k^C(c_n, c_l)$, $\kappa_k^C(\mathbf{c}_n, \mathbf{c}_l) = \kappa_k^C(\mathbf{c}_n - \mathbf{c}_l, \mathbf{c}_n - \mathbf{c}_l) = \kappa_k^C(\mathbf{g}, \mathbf{h}) \kappa_k^C(\mathbf{c}_n, \mathbf{c}_l)$
 - The hyperparameters (scale $s_{k,k}$ and lengthscale $l_{k,k}$) of the GP control the degree of smoothness and the distance over which correlations decay along the covariate.
- **Dynamic Time Warping for Alignment:**
When covariates (e.g., developmental times) are misaligned across groups (such as different species with non-identical developmental stages), MEFISTO incorporates an alignment step based on dynamic time warping (DTW). This data-driven alignment finds a non-linear warping function that minimizes differences in the latent space between groups, effectively mapping the covariate values onto a common scale.
- **Variational Inference:**
To infer the latent factors, weights, and GP hyperparameters, MEFISTO employs variational inference that optimizes a lower bound on the model evidence. The framework is designed to scale to large datasets by leveraging sparse GP approximations and, when possible, exploiting Kronecker product structures in the covariance matrices.

3. Experiments and Applications

The paper validates MEFISTO through a series of experiments, demonstrating its advantages over conventional methods that ignore covariate structure.

Simulated Data

- **Purpose:**
To benchmark factor recovery and imputation performance.
- **Setup:**
 - Multiple simulated datasets were generated with varying numbers of time points, groups, and levels of missing data.
 - Three types of latent factors were simulated: one non-smooth, one smooth but group-specific, and one smooth and shared across groups.
- **Results:**
MEFISTO outperformed traditional factor analysis methods (e.g., MOFA without temporal awareness) in recovering the true latent space and in the accurate imputation of missing values, especially when data were sparsely sampled.

Evolutionary Atlas of Organ Development

- **Dataset:**
Gene expression data from 5 species (groups) and 5 organs (views) across 14–23 developmental time points.
- **Application:**
MEFISTO was used to infer 5 latent factors capturing developmental programs.
- **Findings:**
 - **Shared vs. Group-Specific Factors:**
The first three factors showed similar temporal profiles across all species, reflecting conserved developmental programs. In contrast, factors 4 and 5 captured species-specific variations (e.g., divergence in human or opossum trajectories).
 - **Temporal Alignment:**
Despite missing data and non-aligned developmental stages, MEFISTO successfully learned a common developmental timeline and aligned the data across species.
 - **Interpretation:**
Analysis of factor weights revealed organ-specific gene signatures (e.g., upregulation of GFAP in brain tissues) and broad developmental processes (e.g., cell cycle regulation).

Longitudinal Microbiome Study

- **Dataset:**
Microbiome profiles from infants collected over the first 2 years of life, characterized by extremely sparse data (over 90% zeros) and many missing time points.
- **Application:**
MEFISTO extracted latent factors that capture temporal trajectories in the infant gut microbiome.
- **Findings:**
 - **Factor Associations:**
One factor was associated with delivery mode (with vaginal delivery linked to higher abundance of genera like *Bacteroides* and *Faecalibacterium*), while another captured variation related to infant diet.
 - **Imputation and Interpolation:**
The model demonstrated robust performance in imputing missing values and interpolating factor trajectories to unobserved time points.

Single-Cell Multi-Omics and Spatial Transcriptomics

- **Single-Cell Multi-Omics:**
MEFISTO was applied to scNMT-seq data (integrating RNA expression, DNA methylation, and chromatin accessibility) from early mouse development. The model was able to:

- Jointly capture coordinated epigenetic and transcriptional changes.
 - Denoise data and impute missing molecular layers.
 - **Spatial Transcriptomics:**
The method was further applied to spatially resolved transcriptomics data from the mouse brain. MEFISTO identified spatial patterns that correspond to anatomical structures (e.g., choroid plexus) without requiring single-cell reference data.
 - **Benefits:**
These applications underscore MEFISTO's versatility in handling both temporal and spatial covariates across various omics platforms.
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4. How MEFISTO Works

- **Input:**
The method takes as input a set of data matrices (each representing a different view) for multiple groups of samples. Each sample is annotated with continuous covariate values (time, space).
- **Latent Factor Decomposition:**
The data are decomposed into a set of latent factors (captured in matrix $Z \in \mathbb{R}^{n \times g}$) and weight matrices ($W \in \mathbb{R}^{m \times g}$) that linearly combine to approximate the original data.
- **Modeling Covariate Dependency:**
Each latent factor is assumed to be generated by a Gaussian process over the covariate. This allows the model to impose smoothness constraints, meaning that factor values change gradually with time or space.
- **Alignment:**
For datasets with misaligned covariates (e.g., developmental stages in different species), dynamic time warping is used to align the covariate values to a common timeline.
- **Inference:**
Variational inference is used to approximate the posterior distributions of the latent variables and hyperparameters. Sparse GP approximations and, if available, Kronecker product decompositions, help reduce computational complexity.
- **Downstream Analyses:**
Once trained, MEFISTO enables:
 - **Imputation:** Estimating missing data values based on the latent factors.
 - **Interpolation/Extrapolation:** Predicting latent factor values at new covariate points.
 - **Interpretation:** Analysis of factor weights to uncover biological signatures, as well as assessing smoothness and sharedness of the factors to understand the underlying variation.

5. Conclusion

MEFISTO represents a significant advancement in the analysis of high-dimensional, multimodal biological data by integrating continuous covariate information into factor analysis. It not only recovers interpretable latent factors that capture smooth temporal or spatial variation but also aligns and integrates data across heterogeneous groups. The extensive experiments in the paper—from simulations to diverse biological applications—demonstrate its superior performance in imputation, interpolation, and uncovering biologically meaningful patterns compared to conventional methods.